

I develop UniversalToolchain/Wist2, a typed language composition and runtime SDK with cross-package plans, exact manifest binding, callable-first SSA, the public Wist facade, and experimental PlanFuzz.

Typed composition plans

routes, passes, capabilities, exact executors

Callable-first SSA

descriptor semantics and verifier-gated roundtrip

1,465 / 1,465

0 failures; nine packages and clean consumers

EXPERIENCE

- 2024 — PRESENT

UniversalToolchain / Wist2

 - Typed composition/runtime SDK for independent language packages, deterministic routes/pass ordering, and schema-v5 lock identity.
 - Fail-closed runtime policies, exact manifest binding, and lifecycle ownership.
 - Callable-first SSA with descriptor semantics, effects, determinism, trust, and a verifier-gated roundtrip.
 - Experimental PlanFuzz with a language-neutral core, an Acme adapter, a restricted Wist Int32 adapter, seven oracle families, replay, and reducers.
- JULY 1 — AUGUST 31, 2026

MCST · LLVM · 0.25 FTE

 - 0.25 FTE: graph algorithms, compiler construction, and LLVM.
 - Developing an LLVM optimization pass.

SELECTED PROJECTS

- PlanFuzz

Experimental configuration-aware differential testing with a language-neutral core, Acme and restricted Wist Int32 adapters, fresh-process confirmation, and deterministic program/plan reduction.
- PS-form Analyzer

Ranked 1st of 49 with 104/104 tests; conservative memory-dependence analysis and an exact-oracle harness.
- x86-64 Codegen Lab

SSA-like IR, liveness, linear scan, iced-x86 emission, and differential validation.

SKILLS

- Language architecture

typed artifacts, contribution graphs, routes, passes, manifests
- IR / SSA

Bytecode/AIR, callable descriptors, CFG, dominance, folding, SCCP-lite, DCE
- Verification

differential/metamorphic testing, exact oracles, replay, reducers, parity
- Languages

C#/.NET, C++23, C17, Python, Rust

EDUCATION

Software Engineering student at HSE University

RECOGNITION

HSE Vysshaya Proba prize-winner; top MEPhI Junior results in 2025/2026; Baltic Grand Prize.

TECHNICAL COMMUNICATION

About 50 students; practical NASM IA-32 course; public documentation and technical writing.

AVAILABILITY

Moscow / remote · up to 20 hours/week from September 2026